

Computer Vision 2026 Final Project

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1 Video BD-rate

id	Video Name	BD-rate
1	AMS05	-37.123%
2	WalkInPark	-26.361%
3	Procession	-22.909%
4	Zombie	-13.721%
Average of first 4 videos		-25.029
5	AMS08	-36.135%
6	FootBall	-19.775%
Average of all 6 videos		-26.004%

Table 1: BD-rate of all videos

2 Problem Definition

Given a number of video-frame T , a sequence of high-resolution even frames $I_{Even}^h = \{\hat{E}_t\}_{t=0, t \equiv 0 \pmod 2}^T$ and a sequence of low-resolution frames $I_{Odd}^h = \{B_t\}_{t=1, t \equiv 1 \pmod 2}^T$, a sequence of high-resolution odd frames groundtruth $I_{Odd}^h = \{\hat{B}_{2t+1}\}_{t=0, t \equiv 1 \pmod 2}^T$

our goal is to find a function f which can upscale the low resolution image B_t to high-resolution frame $\hat{B}_t$ by the information in B_t and the neighbor high-resolution frame $\hat{E}_{t-1}, \hat{E}_{t+1}$. Beside, we wish the the MACs/Pixel not greater than 200k.

In other word, we want to solve the following problem:

$$\min_{\theta} \sum_{t=1, t \equiv 1 \pmod 2}^T \|f_{\theta}(B_t, \hat{E}_{t-1}, \hat{E}_{t+1}) - \hat{B}_t\|^2$$

w.r.t. MACs_per_pixel(f_{θ}) < 200k

3 Method

3.1 Baseline: Simple Upscaling

This baseline directly upsamples the FHD base layer to 4K without using temporal information from adjacent frames. Although this method preserves the global structure, it lacks high-frequency details and usually produces blurry results.

3.2 Optical Flow Warping

3.2.1 Introduction

To utilize temporal information, optical flow method is used to estimate optical flow between the base frame and adjacent frames. The previous and next 4K frames are then warped according to the estimated motion field, so their details can be aligned with the target frame.

In traditional methods like Farneback and Lucas Kanade, the optical flow is inconsistent under quick motion and luminance variety. In our method, we chose a Deep Learning model called RAFT(Recurrent All-Pairs Field Transforms) to deal with this problem. First, the model transforms the images into feature maps, and then performs All-pairs correlation between two images. The model then predicts the optical flow with recurrent updates, to ensure the precision.

3.2.2 Compute with efficiency

Given two FHD frames, Image 1 and Image 2, we first downsample the images(In our method, we downsample the image into $\frac{2}{3}$ of its origin), then we input the two downsampled images into the RAFT model and retrieve the optical flow map. Finally, we resize the map back to FHD.

3.2.3 Warping

Our method performs backward warping on a single image plane using an optical flow field. It first creates a coordinate grid for all pixels, then adds the flow displacement to get the source sampling positions. `cv2.remap()` uses these positions to sample pixels from the input plane with bilinear interpolation, producing the warped result.

The function also computes a valid mask by checking whether each sampling position is inside the image boundary. Pixels whose mapped coordinates go outside the image are marked as invalid and filled with 0 by `cv2.remap()`. In this project, it is used to align previous or next frames to the target frame before fusion.

3.3 Fusion Strategy

After warping, a fusion module is applied to combine the upscaled base frame and the warped neighboring frames. The simplest approach is weighted averaging, where the final result is computed as a linear combination of the base, warped previous frame, and warped next frame.

3.3.1 Adaptive Fusion

The adaptive fusion calculate the temporal weights with per-pixel weights that reflect the reliability of each warped frame. For each pixel, the residual error between the warped frame and the base frame is used to estimate how well the warping has aligned the temporal information:

$$e_p = |\hat{E}_{t-1} - B_t|, \quad e_n = |\hat{E}_{t+1} - B_t| \quad (1)$$

These residual errors are converted into reliability weights using a Gaussian decay:

$$w_p = \exp\left(-\frac{e_p}{\sigma}\right), \quad w_n = \exp\left(-\frac{e_n}{\sigma}\right) \quad (2)$$

where σ is a bandwidth parameter (set to 30 in our experiments). A smaller residual error yields a larger weight, meaning the fused result trusts the warped frame more when it closely matches the base.

The validity masks are then applied to suppress contributions from out-of-boundary warped pixels:

$$w_p \leftarrow w_p \cdot \mathbf{m}_p, \quad w_n \leftarrow w_n \cdot \mathbf{m}_n \quad (3)$$

Additionally, a prev-next consistency check is applied. When both temporal frames are valid, their mutual disagreement is penalized:

$$c = \exp\left(-\frac{|\hat{E}_{t-1} - \hat{E}_{t+1}|}{\sigma}\right), \quad w_p \leftarrow w_p \cdot c, \quad w_n \leftarrow w_n \cdot c \quad (4)$$

This suppresses regions where the two warped frames contradict each other, which typically indicates inaccurate optical flow estimation.

The base frame is assigned a fixed minimum weight $w_b = 0.3$ to ensure stability. The final output is:

$$P = \frac{w_b \cdot B_t + w_p \cdot \hat{E}_{t-1} + w_n \cdot \hat{E}_{t+1}}{w_b + w_p + w_n + \epsilon} \quad (5)$$

where $\epsilon = 10^{-6}$ avoids division by zero.

3.4 Super Resolution Module

In addition to temporal warping, a lightweight EDSR-based super-resolution model is trained on the Y channel. Since the Y channel contains most structural and texture information, the model focuses on enhancing luminance details while keeping chroma processing simpler.

3.5 Memory and Computational Cost

The SR model used in our pipeline is **EDSRSmall** with $C = 48$ feature channels, $N = 8$ residual blocks, and $\times 2$ upscaling scale. The total number of trainable parameters is **437 K**, requiring approximately **1.67 MB** of storage in 32-bit float precision.

The multiply-accumulate operations (MACs) are computed per layer for a single FHD (1920×1080) Y-channel input. For a convolution layer with C_{in} input channels, C_{out} output channels, kernel size $K \times K$, and spatial resolution $H \times W$, the MACs are:

$$\text{MACs} = C_{\text{in}} \times C_{\text{out}} \times K^2 \times H \times W \quad (6)$$

Layer	C_{in}	C_{out}	Resolution	MACs
Head (Conv 3×3)	1	48	1920×1080	0.90 G
Body: $8 \times \text{ResBlock}$ ($2 \times \text{Conv}$)	48	48	1920×1080	688.0 G
Body tail (Conv 3×3)	48	48	1920×1080	43.0 G
Upsample (Conv 3×3)	48	192	1920×1080	172.0 G
Tail (Conv 3×3)	48	1	3840×2160	3.58 G
Total				907.4 G

Table 2: MACs breakdown of EDSRSmall ($C = 48$, $N = 8$, $\times 2$) for one FHD Y-channel frame.

The total MACs for one frame amounts to **907.4 GMACs**. Normalised by the number of output 4K pixels ($3840 \times 2160 = 8,294,400$), this yields approximately **109.4 K MACs per output pixel**. The dominant cost is the 8-block residual body operating at full FHD resolution, which accounts for over 75% of total computation. The chroma channels (U, V) are upsampled using bicubic interpolation and incur negligible additional cost.

4 Experiments

We perform experiments of different upscale method, optical flow method and fusion methods, checking the necessity of each stages.

4.1 Experimental Setup

In the experiments, we only calculate the BD rate of first 4 public files, excluding the hidden videos. The input consists of a low-resolution FHD base frame and its neighboring 4K frames. The goal is to reconstruct the target 4K frame. The reconstructed result is compared with the original 4K ground truth frame.

4.2 Compared Methods

Following is the average BD-rate of different methods:

Experiments	Upscale Method	Optical Flow	Fusion Method	Average BD-rate
Exp. 1	Simple Resize	Unused	Unused	139.8%
Exp. 2	Simple Resize	RAFT	Weighted sum	47.8%
Exp. 3	Y-Only EDSR	RAFT	Weighted sum	22.3%
Exp. 4	Y-Only EDSR	RAFT	Y-only Adaptive fusion	-12.1%
Exp. 5	Y-Only EDSR	RAFT	Adaptive fusion for all channel	-25.1%

As we can see, the optimal pipeline will use all steps we mentioned above, which prove the necessity and the importance of all steps.

5 Discussion

For our super-resolution based project, the keypoint is how to use the the information of previous and next frame to enhance the upsampled image. Conceptually, we need to use the base image to cache the sketch of the image, and use adjacent frames to enhance the details.

Our method implemented this idea by calculating optical flow, warping image and linearly adding super-resolution model output and the warped high resolution frames. Such method is effective but naive, the linear adding will make ghosting at the output image.



(a) Original



(b) Model Output

Figure 1: Comparison between original discussion graph and model output.

As picture showed above, the contour of the figure become unclear.

To fix this issue, we tried to use CNN as the fusion layer or directly input the high resolution pictures into EDSR model but both method failed. The change in the input or the fusion layer causing the performance drop sharply.

We think it may difficult to fuse all information by only EDSR model or small layer CNN model, maybe the structure of model needed to be adjust or the number of coefficients is needed. In total, for our super-resolution based framework, the adaptive linearly adding is still the optimal choice for fusing informations.